Aleksandar Dubljevic

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Education

University of Waterloo & Wilfrid Laurier University — Expected May 2028
Bachelor of Computer Science and Bachelor of Business Administration Double Degree — *Waterloo, ON*
Awards: Pearl Sullivan Hack the North Scholarship | Relevant Coursework: Data Structures and Algorithms, Operating Systems

Experience

AI Engineer — Jan 2026 – Apr 2026; Sept 2026 – Present
RBC Capital Markets — *Toronto, ON; New York, NY*

- Architected a text-to-SQL **MCP** server enabling **35+ traders** to query **410M+ rows** of live and historical trade data in natural language, with insights supporting **$118M+** in trades during its first week in production
- Built **MLflow** evaluation pipelines for an LLM news retrieval system used by **650+ employees**, enabling benchmarking that identified failure modes and validated fixes that improved accuracy from **35% to 95%**
- Partnered with traders, engineers, database owners, and compliance to deploy MCP services with **Docker, Helm, and Kubernetes**, adding **RBAC and audit logging** for secure use across multiple desks
- Developed AI agents using the **Claude Agent SDK**, creating a shared library of reusable skills and sub-agents that automated multi-step workflows, increasing median monthly AI usage from **7 to 15 days per user**

Cloud Engineer — Jan 2025 – Apr 2025
Canadian Imperial Bank of Commerce — *Toronto, ON*

- Deployed and tested **Azure** resource configurations (Databricks, VMs, Storage) for enterprise applications across dev/test/prod environments, completing **63 Jira tickets** with zero post-deployment rollback incidents as part of an Agile Scrum team
- Resolved a **SEV2** outage impacting critical banking services by tracing network flows with **Azure Network Watcher** to identify misconfigured firewall rules, collaborating with senior engineers to restore connectivity for thousands of users within 2 hours
- Automated Azure resource deployments by executing **Terraform** scripts through **Azure DevOps CI/CD** pipelines, reducing manual provisioning time by **42%** while ensuring version-controlled, consistent infrastructure across environments

Software Engineer - Dashboard Lead — Sep 2023 – Dec 2025
University of Waterloo Formula Electric — *Waterloo, ON*

- Led the development of an internal **Grafana** dashboard for live telemetry, enabling team members to monitor real-time performance from over **150 data sources**, eliminating the need to restart the vehicle to view sensor data
- Built a **Debian Linux** heads-up display, implementing real-time CAN bus message processing on an embedded system
- Built HIL safety tests for precharge and discharge control logic, validating high-voltage embedded systems

Projects

LLM RL Training (Qwen3-4B) | *Python, Tinker, vLLM, Hugging Face*

- Trained a 4B parameter Qwen model to **99%+ solve accuracy** on NYT Connections puzzles via supervised fine-tuning (LoRA) followed by GRPO-based **reinforcement learning**
- Designed a **verifiable reward scheme** around objectively checkable groupings, incentivizing correctness and token efficiency
- Deployed the model locally using **vLLM**, constraining context size and leveraging prefix caching to reduce redundant computation during multi-turn inference

Camp | **HackMIT 2026 Ramp Track Winner** | *Swift, Python, FastAPI, PostgreSQL*

- Integrated a **Jev-powered recommendation engine** into native Mac and iPhone apps, personalizing meal suggestions through **natural-language feedback**, ratings, and order history
- Engineered cross-device session synchronization with Swift concurrency, **FastAPI**, and **PostgreSQL**, using optimistic concurrency control and automatic conflict reconciliation to prevent stale updates across macOS and iOS

Dime Defender | **Hack the North 2024 Finalist** | *AWS, ElevenLabs, Voiceflow, OpenAI API*

- **Top 12** among **800+ participants** at Hack the North 2024 for developing an AI-powered Chrome extension that generated contextual spending interventions during e-commerce checkout
- Architected a scalable **AWS** serverless backend using Lambda and API Gateway that supported over **75 concurrent users**, achieving **sub-500ms** response times through API request optimizations including pagination and payload compression

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, Shell Script, Go, Rust, Swift
AI/ML & Frameworks: MLflow, Hugging Face, vLLM, LangChain, LlamaIndex, Flask, React, Node.js, Next.js
Cloud & Developer Tools: Azure, AWS, Kubernetes, Docker, Terraform, GitHub Actions, Git, GitHub, Claude Code, Codex